

Dr. André Bauer

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Curriculum Vitae

RESEARCH INTEREST

My research is in performance engineering for AI and cloud systems: I make the infrastructure behind modern AI efficient, predictable, and trustworthy by combining systems measurement and benchmarking with forecasting and machine learning.

ACADEMIC CAREER

University of Chicago Booth School of Business

VISITING ASSISTANT PROFESSOR OF ECONOMETRICS AND STATISTICS

Chicago, United States

since September 2026

Illinois Institute of Technology

ASSISTANT PROFESSOR

Chicago, United States

August 2024 — August 2026

Argonne National Laboratory

ASSOCIATED RESEARCHER

Lemont, United States

January 2023 — August 2024

University of Chicago

POSTDOCTORAL SCHOLAR (PROF. IAN FOSTER)

Chicago, United States

November 2022 — August 2024

University of Würzburg

RESEARCH GROUP LEADER (PROF. SAMUEL KOUNEV)

Würzburg, Germany

January 2021 — October 2022

University of Würzburg

RESEARCH ASSISTANT

Würzburg, Germany

October 2016 — December 2020

SCHOOL & ACADEMIC EDUCATION

University of Würzburg

DR. RER. NAT. (COMPUTER SCIENCE)

Würzburg, Germany

October 2016 — December 2020

- **Dissertation** “Automated Hybrid Time Series Forecasting: Design, Benchmarking, and Use Cases”
- **Advisor** Prof. Dr. Samuel Kounev
- **Grade** Summa cum laude (1.0) — Defense December 22, 2020

University of Würzburg

MASTER OF SCIENCE IN COMPUTER SCIENCE

Würzburg, Germany

October 2014 — September 2016

Blekinge Institute of Technology

SEMESTER ABROAD (ERASMUS PROGRAM)

Karlskrona, Sweden

August 2015 — January 2016

University of Würzburg

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Würzburg, Germany

October 2012 — September 2014

AWARDS & SCHOLARSHIPS

Best Industry Paper Award, ACM/SPEC International Conference on Performance Engineering (ICPE)

Florence, Italy

For the paper “Evaluating Kubernetes Performance for GenAI Inference: From Automatic Speech Recognition to LLM Summarization”.

May 2026

Best Student Paper Award, IEEE International Conference on e-Science (eScience)

Chicago, United States

For the paper “Efficient Privacy-Preserving Recommendation on Sparse Data using Fully Homomorphic Encryption” (co-author).

September 2025

Best Teacher Award, Illinois Institute of Technology

Chicago, United States

The department of computer science honors the best teacher every year.

April 2025

GAIN Participation Grant, German Academic International Network

Bonn, Germany

The GAIN conference connects internationally mobile German scientists with representatives from the scientific community, politics, and industry.

May 2023

Runner-up for Best Data Challenge Paper , ACM/SPEC International Conference on Performance Engineering (ICPE) For the paper “Efficient Data Processing: Assessing the Performance of Different Programming Languages”.	Coimbra, Portugal April 2023
2023 SPECTacular Award , Standard Performance Evaluation Corporation (SPEC) The SPECTacular Award is given to individuals who have significantly exceeded expectations in their contributions to SPEC.	Gainesville, United States February 2023
Leadership Academy Fellowship , German Scholars Organization The Leadership Academy Fellowship is awarded to 25 outstanding researchers from all disciplines.	Berlin, Germany January 2023
Best Industry Paper Award , ACM/SPEC International Conference on Performance Engineering (ICPE) For the paper “Why Is It Not Solved Yet? Challenges for Production-Ready Autoscaling”.	Virtual conference April 2022
SPEC Kaivalya Dixit Distinguished Dissertation Award , SPEC The SPEC Kaivalya Dixit Distinguished Dissertation Award is an annual award recognizing one outstanding doctoral dissertation within the scope of the SPEC Research Group in terms of scientific originality, scientific significance, practical relevance, impact, and presentation.	Gainesville, United States April 2022
Distinguished Dissertation Award , Faculty of Mathematics and Computer Science The Distinguished Dissertation Award is an annual award for outstanding dissertations from the Faculty of Mathematics and Computer Science.	Würzburg, Germany July 2021
Dissertation Award , Unterfränkische Gedenkjahrstiftung für Wissenschaft The Dissertation Award is an annual award for outstanding dissertations from all disciplines.	Würzburg, Germany May 2021
Summa cum laude , University of Würzburg Grading of the doctoral thesis ‘Automated Hybrid Time Series Forecasting: Design, Benchmarking, and Use Cases’.	Würzburg, Germany December 2020
Winner of the FameLab State of Bavaria final , British Council In this science communication competition, researchers must present their work in three minutes in an understandable and humorous way.	Regensburg, Germany March 2020
Sokrates/Erasmus Grant , University of Würzburg Mobility grant for a semester abroad in Sweden.	Würzburg, Germany August 2015 — January 2016
Deutschlandstipendium , University of Würzburg The Deutschlandstipendium supports talented and high-achieving students, taking into account not only first-class grades but also social commitment and special personal achievements.	Würzburg, Germany October 2014 — September 2015
Heinrich-Thein Award , Heinrich-Thein-Stiftung Award for talented students pursuing the “second-chance education”.	Haßfurt, Germany November 2009

TEACHING ACTIVITIES

Analysis of Financial Time Series (BUSN 41921) , University of Chicago Booth School of Business Lecturer of this MBA course (English), Spring Quarter 2027.	Chicago, United States March 2027 — June 2027
Advanced Business Statistics (BUSN 41001) , University of Chicago Booth School of Business Lecturer of this MBA course (English), Winter Quarter 2027.	Chicago, United States January 2027 — March 2027
Time Series Analysis and Forecasting , Illinois Institute of Technology Lecturer of this self-created master’s course (English).	Chicago, United States January 2026 — May 2026
Operating Systems , Illinois Institute of Technology Lecturer of this bachelor’s and master’s course (English).	Chicago, United States August 2025 — December 2025
Time Series Analysis and Forecasting , Illinois Institute of Technology Lecturer of this self-created master’s course (English).	Chicago, United States January 2025 — May 2025

Distributed Systems, University of Chicago

Co-Lecturer with Prof. Kyle Chard of this master's course (English).

Chicago, United States

September 2023 — December 2023

Time Series Analysis and Forecasting, University of Würzburg

Lecturer of this self-created master's course (English).

Würzburg, Germany

April 2022 — September 2022

Advanced Programming, University of Würzburg

Co-Lecturer with Prof. Samuel Kounev of this master's course (English).

Würzburg, Germany

October 2021 — March 2022

Computer Science: Getting Started, University of Würzburg

Co-Lecturer with Prof. Samuel Kounev of this bachelor's course (German).

Würzburg, Germany

April 2021 — September 2021

Operating Systems, University of Würzburg

Co-Lecturer with Dr. Nikolas Herbst of this bachelor's/master's course (German).

Würzburg, Germany

October 2020 — March 2021

Software-Engineering Seminar, University of Würzburg

Instructor of this bachelor's/master's course (German).

Würzburg, Germany

October 2016 — March 2021

Operating Systems (Exercise), University of Würzburg

Coordinator & instructor of this bachelor's/master's course (German).

Würzburg, Germany

October 2016 — March 2021

Practical Course in Software, University of Würzburg

Coordinator & instructor of this bachelor's course (German).

Würzburg, Germany

October 2016 — September 2020

SCIENCE COMMUNICATION COMPETITIONS**1st South Side Science Festival Science Slam**

UNIVERSITY OF CHICAGO

Chicago, USA

September 2023

6th GAIN Science Slam

GERMAN ACADEMIC INTERNATIONAL NETWORK

Boston, USA

August 2023

21st Stuttgart Science Slam

LUUPS

Stuttgart, Germany

October 2022

FameLab State of Bavaria final

BRITISH COUNCIL

Regensburg, Germany

March 2021

6th Alumni-Science Slam

UNIVERSITY OF WÜRZBURG

Würzburg, Germany

November 2020

FameLab German final

BRITISH COUNCIL

Bielefeld, Germany

May 2020

FameLab State of Bavaria final

BRITISH COUNCIL

Regensburg, Germany

March 2020

14th Munich Science Slam

LUUPS

Munich, Germany

February 2020

33rd Dortmund Science Slam

LUUPS

Dortmund, Germany

November 2019

INVITED TALKS**An Open Function-as-a-Service Dataset From the Edge to the Cloud**

PARSL FEST

Chicago, US

September 2024

Academic Success: Navigating your Journey

UNIVERSITY OF CHICAGO

Chicago, US

February 2024

Smart Infrastructure Management: Navigating Efficiency in Scientific Computing

ILLINOIS INSTITUTE OF TECHNOLOGY

Chicago, US

November 2023

Why Auto-Scaling Still Matters

DEPAUL UNIVERSITY

Chicago, US

October 2023

Towards an Intelligent Infrastructure for Data Science Clouds

CHALMERS UNIVERSITY OF TECHNOLOGY

Edmonton, Canada

May 2023

Autonomous Data Analytic Clouds

CHALMERS UNIVERSITY OF TECHNOLOGY

Gothenburg, Sweden

February 2023

A Short Overview of Auto-Scaling & Why It Is Not Yet Solved

MONGODB

Measuring Energy-Efficiency in Digital Systems

HASSO-PLATTNER-INSTITUT

Virtual meeting

April 2022

Potsdam, Germany

March 2022

FUNDING

NSF CRII Grant (\$150,000), National Science Foundation

August 2025

PI for the project "Co-Simulation-Based Optimization of Container Orchestration Platforms".

Google Research Grant (\$5,000), Google

May 2025

Acquisition of Google Cloud credits for the project "Assessing Microservice Applications Performance for Benchmarking".

Google Research Grant (\$1,000), Google

March 2023

Acquisition of Google Cloud credits for the project "Performance Analysis and Prediction of Microservices in Cloud Environments".

DFG Research Grant (\$347,235), German Research Foundation

January 2023

PI for the project "MINERVA: Performance Prediction of Microservice Applications with Black-box Container Orchestration".

Industrial project, DATEV eG

May 2022

Assistance for the extension of the project "Self-Configuring Autoscaler" (\$30,068), with a project duration of 3 months.

Industrial project, DATEV eG

July 2021

Assistance with the application of the project "Self-Configuring Autoscaler" (\$71,200), with a project duration of 6 months.

Normalverfahren Grant 2021 (\$3,234), Universitätsbund Würzburg

April 2021

Acquisition of the Normalverfahren Grant for the project "Prediction of Winter Mortality of Bee Colonies Using Machine Learning".

ACADEMIC SERVICE

Organization Committee

- Workshop on Hot Topics in Cloud Computing Performance 2025
- International Conference on Performance Engineering (Data Challenge) 2025
- International Conference on e-Science (Posters) 2024
- International Conference on Performance Engineering (Posters and Demos) 2024
- International Workshop on Performance-Data Analytics and Data-Management 2023, 2022

Program Committee

- International Conference on Utility and Cloud Computing 2026, 2025, 2024, 2023, 2022
- International Conference on Time Series and Forecasting 2026, 2025, 2024, 2023, 2022
- International Conference on Data Science, Technology and Applications 2026, 2025, 2024, 2023
- Int'l Conference on Autonomic Computing and Self-Organizing Systems 2026, 2025, 2024
- International Conference on e-Science 2026, 2025, 2024
- International Conference on Performance Engineering 2027, 2026, 2025, 2024
- International Workshop on Intelligent and Adaptive Edge-Cloud Operations and Services 2026, 2025
- International Conference for High Performance Computing, Networking, Storage and Analysis 2026
- International Conference on Software Engineering for Adaptive and Self-Managing Systems 2027, 2026
- Workshop on Hot Topics in Cloud Computing Performance 2026
- Symposium on Software Performance 2025, 2024, 2023, 2022
- International Symposium on Cluster, Cloud and Internet Computing (Artifact Evaluation) 2025, 2024
- International Conference on Service-Oriented System Engineering 2025
- International Conference on Cloud Computing and Services Science 2024
- International Symposium on Cluster, Cloud and Internet Computing 2023
- Int'l Conference on Autonomic Computing and Self-Organizing Systems (Posters and Demos) 2023, 2022
- International Conference on Performance Engineering (Artifact Evaluation) 2020
- Workshop on Evaluations and Measurements in Self-Aware Computing Systems 2019

Associate Editor

- Frontiers in Digital Health, Health Informatics since March 2026

Editorial Board Member

- ICCK Journal of Systems Scalability
- MDPI Data

since October 2025
since January 2025

Financial Chair

- International Conference on e-Science

2025

Publicity Chair

- International Conference on Performance Engineering
- Workshop on Evaluations and Measurements in Self-Aware Computing Systems

2020
2019

Standard Performance Evaluation Corporation

- Founder and chair of the SPEC RG Predictive Data Analytics Working Group
- Elected editor of the annual SPEC RG newsletter
- Elected release manager of the SPEC RG Cloud Working Group

since June 2021
April 2017 — May 2023
October 2018 — May 2021

Journal Reviews

- Journal of Systems & Software
- Journal of Parallel and Distributed Computing
- Journal of Grid Computing
- Journal of Systems Scalability
- Performance Evaluation
- Journal of Supercomputing
- Information Processing and Management
- Frontiers Digital Health
- Empirical Software Engineering
- SoftwareX
- Transactions on Knowledge and Data Engineering
- Science of Computer Programming
- Transactions on Autonomous and Adaptive Systems
- Transactions on Cloud Computing
- Transactions on Services Computing
- Journal of Future Generation Computer Systems
- Transactions on Network and Service Management

2026, 2022, 2018
2026, 2021, 2019
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2024, 2023, 2022
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2020
2019
2018

CERTIFICATES

Certificate of Leadership – Smart Skills National Postdoctoral Association

January 2024

Certificate Leadership University of Würzburg

October 2022

Certificate Academic Teaching – State of Bavaria University of Würzburg

November 2020

Certificate Academic Teaching – University of Würzburg University of Würzburg

December 2019

VOLUNTARY ACTIVITIES

Mentoring for high school students

ILLINOIS SCIENCE & TECHNOLOGY COALITION

Chicago, United States

since August 2024

Pen Pal Program

OFFICE OF DIVERSITY AND INCLUSION UNIVERSITY OF CHICAGO

Chicago, United States

February 2023 — August 2024

Mentoring for students

ALUMNI NETWORK UNIVERSITY OF WÜRZBURG

Würzburg, Germany

November 2019 — October 2022

Supervision of scouts

SCHWARZER HAUFEN WÜRZBURG (BDP E.V.)

Würzburg, Germany

July 2019 — February 2020

Volunteer tutoring

NACHBARSCHAFTSHILFE GROMBÜHL

Würzburg, Germany

September 2013 — September 2014

Student council

FRIEDRICH-FISCHER SCHULE

Schweinfurt, Germany

September 2010 — August 2011

Volunteer fire fighter

FFW HORHAUSEN E.V.

Horhausen, Germany

April 2003 — August 2009

LANGUAGES

German Mother tongue

English Fluent